Jinlong (Vincent) Liu

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Education

University of Birmingham Feb 2024 – Present
PhD in Computer Science; research in large language models and natural language generation.

University of Liverpool Sep 2022 – Dec 2023
MSc in Advanced Computer Science.

Central South University of Forestry and Technology Sep 2017 – Jul 2021
BSc in Computer Science and Technology; overall mark: 80%.

Technical Skills

Programming	Python, C, C++.
Libraries	PyTorch, Hugging Face Transformers
LLM fine-tuning	Supervised fine-tuning, LoRA, GRPO, reward modelling
Training systems	DeepSpeed ZeRO-3, PyTorch FSDP, vLLM

Research and Engineering

Doctoral Research Feb 2024 – Present
University of Birmingham, School of Computer Science

Style-Controlled LLM Fine-Tuning *Code*
PyTorch | SFT and GRPO | Distributed training

- Built SFT and GRPO training workflows for an **8B language model** to generate long-form stories in the styles of four classic authors.
- Trained a sentence-transformer style judge with authorship-verification supervision and calibrated its similarity scores into a bounded reward for reinforcement learning.
- Implemented distributed training configurations using FSDP and DeepSpeed ZeRO-3, and integrated a vLLM-served judge for generation-quality scoring.
- Evaluated style transfer across Twain, Austen, Dickens and Hardy; achieved a mean style score of **0.893**, exceeding the evaluated open-weight baselines on this metric. Published at **CoNLL 2026**.

LLM Training Data and Evaluation *Code | Dataset*
LoRA | DeepSpeed | vLLM | Hugging Face Datasets

- Constructed and released a dataset of **263,911 long-form stories**, each paired with scores and synthesised reviews across **14 creative-writing criteria**.
- Built a multi-stage pipeline for story generation, asynchronous LLM evaluation, review synthesis and training-data preparation, with resumable evaluation and dataset-quality checks.
- Implemented LoRA fine-tuning with DeepSpeed ZeRO-3 and vLLM evaluation for **Qwen3-4B, Qwen3-8B and Nemotron-8B**, comparing reasoning-trace supervision with direct-review training.
- Analysed score agreement, report completion and repetition failures; found that direct-review supervision produced more reliable structured reports across all three models.

Additional Projects

Google Research Football Agent Jul 2021 – Sep 2022

PyTorch, GCN, DQN; group project with Prof. Pietro Liò, University of Cambridge

- Collaborated on a football-playing agent combining a graph convolutional network with a Deep Q-Network to learn from interactions between players.
- Represented player and ball positions as 2D coordinates and trained the agent to select from **18 actions** across **four difficulty settings**.
- Contributed to the resulting paper, *Graph Neural Network Based Agent in Google Research Football*, published at **AIAHPC 2022**.

Lung CT Classification Oct 2020 – Jun 2021

TensorFlow, CNN, ResNet; undergraduate dissertation

- Developed a convolutional neural network to classify lung CT scans into **healthy, pneumonia and COVID-19** categories.
- Adapted a ResNet architecture into a simpler model for single-organ CT imagery, investigating a lower-complexity alternative to the standard architecture.

Delivery Route Optimisation Oct 2019 – Jun 2020

Python, genetic algorithms; university student innovation and entrepreneurship competition

- Applied a genetic algorithm to a delivery-planning problem with constraints on employee working hours and vehicle fuel consumption.
- Represented the road network as a graph and updated employee and vehicle states after each delivery stop to support route planning.

Sales and Inventory Modelling Mar 2019 – Jun 2019

National University Students Mathematical Modelling Contest

- Collaborated on a mathematical model for sales and inventory planning, translating the problem requirements into a quantitative formulation.

Experience

Postgraduate Teaching Assistant 2024 – 2026

University of Birmingham; Sep 2024 – Jun 2025 and Sep 2025 – Mar 2026

- Supported students during practical lab sessions, helping them work through exercises and apply course concepts to practical tasks.
- Marked coursework against assessment criteria across two academic years.

EEG Data Processing Oct 2021 – Jun 2022

Shenzhen Institutes of Advanced Technology, Chinese Academy of Sciences

- Processed EEG recordings in **Python and MATLAB** for brain-computer interface research within the Brain Cognition and Brain Disease Institute.
- Applied **principal component analysis (PCA)** and **support vector machines (SVMs)** to analyse data collected during experiments.

Publications

1. **J. Liu** et al. Capturing Classic Authorial Style in Long-Form Story Generation with GRPO Fine-Tuning. **CoNLL 2026.**
2. **J. Liu** et al. When Reasoning Supervision Hurts: TTCW-Based Long-Form Literary Review Generation. **arXiv preprint, 2026.**
3. **J. Liu** et al. Graph Neural Network Based Agent in Google Research Football. **AIAHPC 2022.**

Languages: English, Chinese.